

Akhila Vemuru

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Professional Summary

Data Scientist with 4 years of combined industry and academic experience developing machine learning models, NLP systems, and data analytics pipelines using Python and SQL. Skilled in analyzing large datasets, building predictive models, and creating dashboards to translate complex insights for technical and non-technical audiences. Experience working with NLP, LLM-based systems, and data-driven decision tools to support analytical and operational insights.

Education

University of Houston - Master's in Data Science, GPA: 3.66/4.0 Jan 2024 – Dec 2025

Vidya Jyothi Institute of Technology - Bachelor's in Electronics and Communication, GPA: 8.76/10 Aug 2017 – July 2021

Experience

Data Analyst, Guardian's Embrace – Remote Feb 2026 – Present

- Analyzed **10k+** donor and operational records using Python, SQL, and Excel to generate insights supporting outreach and fundraising strategies
- Developed interactive Power BI/Tableau dashboards tracking KPIs, improving visibility into donor engagement and program performance by **30%**. Automated reporting workflows using SQL, reducing manual reporting effort by **25%** and improving data accuracy

Data Analyst, University of Houston – Houston, TX Sept 2024 - Present

- Analyzed **500+** HR and financial transaction data using Smartsheet and Excel to automate validation and salary reconciliation, improving audit efficiency by **40%** and communicated statistical insights to students in an accessible manner.
- Analyzed purchasing and financial transaction data for **500+** monthly vouchers in Oracle, performed data cleaning, validation, and trend analysis on **120+** items using Excel to identify trends, support forecasting, and improve decision-making

Data Analyst, Capgemini – Hyderabad, India Aug 2021 – Dec 2023

- Designed SQL ETL workflows to ingest, clean, and transform policy and transaction data into analytics-ready tables.
- Built monitoring dashboards tracking performance across 8+ system integrations, reducing troubleshooting time by **20%**.
- Wrote complex SQL queries for data manipulation, structuring, and query optimization to support operational reporting and performance analysis. Performed data validation and anomaly detection to resolve 60+ production data issues through structured root cause analysis.

Intern, Capgemini – Hyderabad, India Apr 2021 - July 2021

- Developed a vehicle rental platform using Java, Spring Boot, Angular, and SQL, implementing backend services and frontend interfaces to support booking transactions and data storage for **1,000+** test vehicle bookings

Projects

Fraudulent Job Posting Detection and Explanation (NLP, Deep Learning, RAG)

- Built an NLP-based fraud detection system using transformer models (DistilBERT, RoBERTa) to classify **18K+** job postings and detect suspicious patterns in text data.. Improved detection accuracy and interpretability by handling class imbalance with class-weighted loss and Implemented RAG techniques to improve model interpretability and explain prediction outcomes.
- Applied class-imbalance handling and anomaly detection techniques to identify suspicious language patterns, achieving 90%+ ROC-AUC.

House Price Prediction using Machine Learning

- Developed a regression-based predictive model to estimate housing prices using historical property data, achieving **85%** accuracy through regression analysis and feature engineering. Utilized Python for data preprocessing, model training, and evaluation, selecting key features such as location, square footage, and amenities

Holiday Season Sales Analysis and Forecasting | Power BI, DAX, Power Query

- Built interactive Power BI dashboards, Performed forecasting analysis using DAX and time-series forecasting to analyze retail sales trends and support data-driven investment-style decisions.

Disneyland Review Analysis(Python, NLTK, TextBlob, Tableau)

- Reviewed **40,000+** customer reviews to identify common visitor concerns across Disneyland parks using Python NLP Libraries.
- Engineered **45+** features from 6 base fields with Python, built Tableau dashboards showing trends in satisfaction and complaints by year and country.

Technical skills

Languages Tools: Python (Pandas, NumPy, Scikit-learn), SQL, Excel (Advanced)

Analytics: Data Analysis, EDA, Statistical Analysis, KPI Analysis, Forecasting, Anomaly Detection

Visualization: Power BI, Tableau

Data Engineering: ETL Pipelines, Data Cleaning, Data Transformation

Machine Learning: Predictive Modeling, Feature Engineering, NLP, Retrieval-Augmented Generation (RAG)